

MicroGridsPy: A bottom-up, open-source optimization model for multi-year planning of mini-grids in energy-access contexts

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Summary

Many communities in remote regions lack access to modern and sustainable energy services, including reliable electricity. Mini-grids - small local electricity systems built around renewable generation, storage, and backup generators - are one of the main ways to supply them, but designing one means deciding what to build, how large, in what order, and at what cost.

MicroGridsPy is an open-source Python package that supports these decisions. Given electricity-demand and renewable-resource time series together with technical and economic assumptions, it builds and solves an optimization model that determines which generation and storage assets to install, when to expand them, and how to operate them over time. Systems can combine renewable generation, battery storage, backup generators, and an optional grid connection with import and export. The package supports both representative-year and explicit multi-year planning, and can account for uncertain future conditions by optimizing across several scenarios at once.

The optimization core is implemented with Linopy ([Hofmann, 2023](#)) and can use the open-source HiGHS solver ([Huangfu & Hall, 2018](#)) or commercial Gurobi ([Gurobi Optimization, LLC, 2026](#)). MicroGridsPy is available as a Python library, a command-line interface, and an optional Streamlit graphical interface. Projects use structured CSV, YAML, and JSON inputs and store results in self-contained project folders, supporting reproducible and inspectable analyses. The package is openly developed on [GitHub](#) and documented at [Read the Docs](#).

Statement of need

Planning energy access in remote areas requires decisions that couple uncertain and evolving demand with variable locally available renewable resources, high capital costs, reliability requirements, and limited technical data. Planners need to compare renewable generation, storage, and dispatchable backup; account for demand growth and staged investment; and evaluate the cost and reliability consequences of uncertainty. The planning problem is therefore not limited to identifying a least-cost technology mix for a single representative year: the timing of investment, the ageing of assets, and the treatment of unserved demand all shape the result.

Energy-access planning is also broader than technical sizing, as the Comprehensive Energy Solution Planning (CESP) framework sets out in connecting engineering analysis with wider social, contextual, and project-planning considerations ([Colombo et al., 2024](#)). Within that perspective, MicroGridsPy addresses the techno-economic optimization layer: it provides a transparent, bottom-up formulation for system design and operation that jointly represents

40 long-term investment decisions, high-resolution operation, economic costs, carbon-related
41 environmental impacts, the social cost of unmet demand, and uncertainty.

42 The primary users are researchers, energy planners, and practitioners studying rural
43 electrification, mini-grid design, and long-term energy-access strategies who need an open and
44 extensible alternative to proprietary optimization software or to substantial custom model
45 development.

46 State of the field

47 General-purpose energy-system frameworks such as Calliope (Pfenninger & Pickering, 2018),
48 PyPSA (Brown et al., 2018), and OSeMOSYS (Howells et al., 2011) are powerful open-source
49 tools for energy-system optimization across a range of scales and applications. Applying
50 them to detailed mini-grid planning may nevertheless require additional model development to
51 represent features such as battery ageing, generator part-load operation, staged investment
52 under demand growth, lost-load treatment, and scenario-dependent uncertainty.

53 Tools built specifically for off-grid systems are closer to this use case. HOMER (Lambert et
54 al., 2006) is the practitioner standard for hybrid-system sizing but is proprietary, which limits
55 transparency and reproducibility in research. Offgridplanner (Reiner Lemoine Institut, 2026)
56 and CLOVER (Sandwell et al., 2023) are open source and offer strong capabilities in system
57 design, simulation, and spatial distribution planning. MicroGridsPy differs in formulating
58 capacity expansion explicitly as a multi-year optimization problem, with stochastic recourse
59 and mini-grid-specific component representations inside a single economic formulation.

60 MicroGridsPy continues a line of open mini-grid modelling work (Balderrama et al., 2019),
61 but the present version is a substantial redevelopment rather than an incremental release. It
62 contributes (i) a re-architected, installable Python package with a documented API, a command-
63 line interface, and an optional graphical interface, replacing an earlier script-based workflow; (ii)
64 a migration of the optimization core from Pyomo to Linopy; (iii) an annuity-based, dual-rate
65 economic formulation; (iv) explicit power-electronic inverter sizing and a semi-empirical battery
66 degradation model; and (v) typical-year and explicit multi-year formulations sharing a single
67 economic core, with a consistent treatment of stochasticity across both. Together these
68 support studies of technology choice, staged investment under evolving demand, reliability–cost
69 trade-offs, and policy levers such as carbon pricing and renewable-penetration targets.

70 Software design

71 The mathematical model is formulated in Linopy, separating model construction from solver
72 execution and allowing the same formulation to be used from scripts, notebooks, the command
73 line, or the graphical interface. HiGHS provides the open-source solver option, while Gurobi is
74 supported for studies that use a commercial solver.

75 Two planning formulations share the same economic and technical modelling principles. The
76 typical-year formulation represents the system with a recurring representative year and is
77 intended for compact screening studies. The multi-year formulation represents an explicit
78 planning horizon and introduces time-dependent investment decisions, capacity expansion, and
79 the timing of operating and capital costs. This separation is a deliberate trade-off: users can
80 choose a simpler formulation when temporal investment effects are not central, while retaining
81 a common modelling architecture for long-term studies.

82 The model annualizes investment through a capital-recovery factor at each technology's
83 weighted average cost of capital while discounting cash flows at a social discount rate, and can
84 represent technology replacement and remaining asset value in multi-year studies. For storage,
85 it incorporates a semi-empirical degradation model with temperature- and depth-resolved cycle
86 ageing, kept linear through a per-state-of-charge-band marginal-cost formulation. Renewable

87 and battery assets can include explicit inverter sizing, and generator operation can account
88 for partial-load efficiency. These choices add modelling detail where it can materially affect
89 mini-grid design while preserving a tractable linear or mixed-integer formulation.

90 Uncertainty is incorporated within the optimization: investment decisions are separated from
91 scenario-dependent operational recourse, allowing uncertain resource or demand trajectories
92 and other operating parameters to influence both system design and dispatch. Additional
93 constraints and objective terms can represent land availability, renewable-penetration targets,
94 lost-load limits and penalties, and carbon costs.

95 The same core formulation is exposed through the Python API, the command-line interface,
96 and an optional Streamlit workflow that follows the planning sequence of project definition,
97 input generation and validation, optimization, and result inspection. Project data are stored as
98 CSV, YAML, and JSON files in self-contained folders. This imposes some up-front structure
99 on inputs in exchange for reproducibility, explicit configuration, and a clear separation between
100 the modelling layer and the user interface.

101 Research impact statement

102 The modelling approach underlying MicroGridsPy has been applied in peer-reviewed studies of
103 rural electrification, including multi-year sizing under evolving demand (Stevanato et al., 2020),
104 coupling with spatial electrification planning (Peña Balderrama et al., 2020), multi-objective
105 and brownfield mini-grid design (Stevanato et al., 2023), and off-grid planning for a rural
106 community in Nigeria (Agbo et al., 2025); the repository's pubs_list.md records the full
107 publication history. The earlier studies were based on Pyomo implementations of the model,
108 whereas Agbo et al. (Agbo et al., 2025) used an early Linopy-based implementation of the
109 evolving codebase. The package described here consolidates and extends this lineage with the
110 current economic and planning formulations, while providing an installable, documented, and
111 testable Python package.

112 MicroGridsPy is also used in teaching at Politecnico di Milano, in the courses *Engineering and*
113 *Cooperation for Development and Innovative Technologies for Energy*, and in international
114 capacity-building programmes: the Climate Compatible Growth *Energy Modelling Platform*
115 *for Africa*, hosted by the Ghana Institute of Management and Public Administration (Accra,
116 2024), the UN Economic Commission for Africa (Addis Ababa, 2025) and the University of
117 Cape Town (2026); and the European Union–African Union LEAP-RE *RESchools* (Pretoria,
118 2022; Kigali, 2023; Milan, 2024) (LEAP-RE Consortium, 2025).

119 The repository is maintained for continued community use: an automated test suite runs in
120 continuous integration across Python 3.10–3.12 alongside linting and coverage reporting, the
121 package ships runnable end-to-end examples for both planning formulations, and contribution
122 guidelines, a code of conduct, and full API and methodology documentation are maintained
123 alongside the code.

124 AI usage disclosure

125 Generative AI tools were used during the development of MicroGridsPy and in the preparation
126 of this manuscript. Claude (Anthropic, Opus and Sonnet model families) and ChatGPT
127 (OpenAI, GPT-4o and GPT-5 model families) assisted with code generation, refactoring, and
128 test scaffolding during the migration of the codebase from Pyomo to Linopy and its packaging;
129 with documentation and repository and continuous-integration configuration; and with drafting
130 and copy-editing parts of this manuscript.

131 The scientific content of the software was specified by the human authors: the optimization
132 formulation, the economic model, the battery-degradation representation, and the multi-year
133 and stochastic formulations are the authors' own work. The correctness of AI-assisted code was

134 verified by the authors' review of each constraint against the intended mathematical formulation,
135 and by an automated test suite covering the economic, degradation, inverter-sizing, partial-load,
136 and public-API components, which is executed in continuous integration across all supported
137 Python versions. All AI-assisted text was reviewed and edited by the authors, who made every
138 design decision and take full responsibility for the software and for this manuscript.

139 Conflict of interest

140 The authors declare no conflicts of interest.

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148 and Sylvain Quoilin at the Université de Liège (Balderrama et al., 2019), whose formulation
149 underpins the present redevelopment. We thank them, and all contributors listed in the
150 repository's AUTHORS file.

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